

Task 3: Analytics Dashboard & Notification System for LMS Platform

Goal: Enhance the LMS/Subscription platform by adding analytics dashboards and notification systems. This project helps learners transition from CRUD-based systems to data-driven web applications.

Tech Stack:

- Django – Admin panel with analytics and notification management
- FastAPI – User panel backend (activity logging and notifications)
- Celery + Redis (optional) – Background task processing
- Chart.js / Recharts – Data visualization
- MySQL / PostgreSQL – Database

Core Features:

- **Analytics Dashboard (Django):** Display user count, course popularity, monthly revenue, and activity trends.
- **Notifications System:** Email & in-app notifications for plan expiry, course updates, and instructor messages.
- **Activity Logging (FastAPI):** Log user actions such as enrollments and completions.
- **User Panel (FastAPI):** Endpoints to view and manage notifications and activity logs.

Suggested Models:

Model	Fields
Notification	user_id, message, is_read, created_at
ActivityLog	user_id, action_type, action_detail, created_at
AnalyticsRecord	date, total_users, active_subscriptions, revenue, popular_course

APIs to Implement:

Endpoint	Method	Description
/notifications/	GET	Fetch user notifications
/notifications/mark-read/	POST	Mark notification as read
/activity/	POST	Log user actions
/analytics/overview/	GET	Fetch admin analytics summary
/analytics/monthly/	GET	Fetch monthly revenue stats

Key Learning Objectives:

- Implement real-time and asynchronous notifications
- Use Celery for background email or analytics tasks
- Work with aggregation queries and analytics visualization
- Integrate Chart.js with Django templates

Deliverables:

- Source code uploaded to GitHub
- README.md with setup and API documentation
- Postman collection for all endpoints
- Screenshots or short demo video showing analytics dashboard and notifications

Evaluation Criteria:

Area	Weight
Code Quality & Structure	25%
Analytics Accuracy	25%
Notification Functionality	25%
Documentation & Presentation	25%